LLM Driven Agentic AI Self Improving Service Composition In IoT

Literature Survey

1. Service composition with LLMs and agents

a) Agentic Service-Oriented Computing: A Manifesto for the Next Frontier of Service-Oriented Computing

Year: 2026 | **Domain:** Service-oriented computing, agentic AI

What the research is about: It is a vision paper, not an experimental one. It argues that LLM-based agents introduce a new way of building composite services. The agents can propose, adapt and execute compositions at runtime, instead of relying only on manual specification or formal planning. It then lays out a research agenda. The agenda covers composition models for agentic services, standardized agent service interfaces, QoS-aware composition, and semantic interoperability.

Limitations:

- It proposes questions rather than a working system, so it has no implementation or results to compare against.
- It notes that LLM-based semantic matching is probabilistic and unverified, and it leaves this as an open problem.
- It says QoS-aware composition for agentic services is a major unsolved problem, so it gives no method for it.

b) The IoT Whisperer: A Framework for Intelligent IoT Service Composition through LLMs

Year: 2025 | **Domain:** IoT, service composition, LLM

What the research is about: Setting up IoT workflows by hand is slow and hard to scale. This paper uses the Web of Things framework, Node-RED and LLMs to turn a plain-language description of a system into an executable workflow. The workflow components are generated automatically, which makes IoT development easier for non-experts.

Limitations:

- It is built for generating workflows from descriptions, with no sign of the agent learning from past runs, so there is no self-improvement.
- It is tied to the Node-RED and Web of Things stack, which may limit how far it carries over to other platforms.
- The abstract does not mention QoS optimisation or runtime adaptation. It is worth checking whether the paper tests failures or changing conditions.

c) Leveraging LLMs for Dynamic IoT Systems Generation through Mixed-Initiative Interaction

Year: 2025 | **Domain:** IoT, LLM agents, dynamic service generation

What the research is about: It describes an LLM-agent architecture for IoT systems that change while running. A service discovery step first looks for an existing service that fits a requirement. If none matches, an LLM coding agent generates a new service and adds it to the system's registry and context. The user and the system refine goals together.

Limitations:

- Letting an LLM to generate and deploy new services raises correctness and safety concerns, especially when the services control physical devices.
- The generated service is added to the registry, but I saw no sign that the system learns from earlier successes or failures to get better over time.
- It relies on a user interacting with the system, which may slow down fully autonomous operation.

d) FAME: QoS-aware Async Service Orchestration for Agentic Workflows

Year: 2026 | **Domain:** Agentic workflows, QoS, serverless orchestration

What the research is about: Agents that call tools through MCP (Model Context Protocol) run into latency, cost and context-size problems. This happens when large tool outputs get passed repeatedly through the prompt. FAME treats the input and output of MCP

services as a QoS concern. It uses serverless (FaaS) wrapping, S3 handle passing, output caching and asynchronous callbacks. It evaluates four workflow patterns: ReAct, Reflexion, LLM Compiler and MLZero.

Limitations:

- It focuses on the efficiency of tool invocation, not on choosing or composing the best services for a goal.
- It is built around MCP and serverless infrastructure, so it is not specific to IoT devices.
- Only four workflow patterns are evaluated, so the results may not cover other agent designs.

2. LLM in IoT

a) Large Language Models in the IoT Ecosystem: A Survey on Security Challenges and Applications

Year: 2025 | **Domain:** LLM, IoT, security (survey)

What the research is about: A literature survey of how LLMs are used in IoT. It covers application areas such as smart cities, healthcare and industrial IoT, and the main challenges: security, privacy, latency, and edge computing. It is based on 30 key papers, mostly from 2020 to mid-2024.

Limitations:

- The review is small (30 papers) and stops around mid-2024, so it misses the recent agent-based work.
- The authors acknowledge that very recent or niche works may be missing.
- As a survey it proposes no new method or experiment, so it helps with background but not with your model design.

b) LLMind 2.0: Distributed IoT Automation with Natural Language M2M Communication and Lightweight LLM Agents

Year: 2025 | **Domain:** LLM agents, IoT automation, distributed systems

What the research is about: Having one central LLM control all IoT devices does not scale. LLMind 2.0 keeps a central coordinator that breaks a user's instruction into simple plain-language subtasks. Lightweight LLM agents on each device then write the device-

specific code locally. Experiments with multi-robot warehouses and WiFi networks show better scalability, reliability and privacy than the centralized version.

Limitations:

- Devices need enough capacity to run a lightweight LLM, which many constrained IoT devices cannot do.
- The results come from two scenarios (warehouse robots and WiFi networks), so generalising to other IoT settings is unproven.
- Natural-language messages between devices are flexible but ambiguous. The summary I saw does not say how misunderstandings or conflicts are handled.
- It describes no learning or self-improvement across tasks.

3. Agentic AI

a) Agentic AI: A Comprehensive Survey of Architectures, Applications, and Future Directions

Year: 2026 | **Domain:** Agentic AI, LLM agents (survey)

What the research is about: A survey of how LLM agents are built, from single-loop agents to hierarchical multi-agent systems. It proposes a taxonomy with six parts: Perception, Brain, Planning, Action, Tool Use and Collaboration. It also covers evaluation and open challenges such as hallucinated actions, infinite loops and prompt injection.

Limitations:

- It is a general survey, so it does not address service composition or IoT directly.
- A taxonomy organises existing work, but you would still have to design and test the composition model yourself.
- It names reliability and security problems but does not solve them.

b) Agentic AI: A Comprehensive Survey of Architectures, Applications, and Future Directions

Year: 2025 | **Domain:** Agentic AI, multi-agent systems (survey)

What the research is about: A systematic review of the move from passive AI tools to autonomous systems that pursue goals. It warns against "conceptual retrofitting", meaning treating modern neural agentic systems as if they were old symbolic agents. It covers architectures, multi-agent collaboration, applications and governance.

Limitations:

- It is a broad overview, so detail on any one area, including IoT and service composition, is limited.
- Being a survey, it offers no experiments or system you can reuse.
- The field moves fast, so newer agent work from 2026 is probably not covered.

4. Self-improvement in agents

a) Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents

Year: 2025 (ICLR 2026) | **Domain:** Self-improving agents, coding agents, evolutionary search

What the research is about: The original Gödel machine was a theoretical AI that rewrites itself only after proving the change is beneficial. The Darwin Gödel Machine drops the proof requirement and tests each self-modification on coding benchmarks instead. It keeps an archive of all generated agents and branches new versions from them. Over its run, performance rose from 20% to 50% on SWE-bench and from 14.2% to 30.7% on Polyglot.

Limitations:

- It improves coding agents on coding benchmarks, so it is unproven for other tasks such as service composition or IoT control.
- Self-modification is checked empirically, not proven safe. That is a concern if the agent controls physical devices.
- It needs many benchmark runs, which is costly.
- Coding benchmarks give a clear success signal. IoT composition would need an equivalent feedback signal, such as QoS metrics.

b) A Survey on Self-Evolution of Large Language Models

Year: 2024 | **Domain:** Self-evolving LLMs and agents (survey)

What the research is about: A survey of approaches where LLMs and agents learn from experience they generate themselves. It gives a framework with four stages: acquiring experience, refining it, updating the model, and evaluating the result. It also covers related ideas such as self-instruct, self-play and self-training.

Limitations:

- It was published in 2024, so it predates much of the recent self-improving agent work.
- It centres on LLMs and general agents, with little on IoT or service composition.
- It is a survey, so it gives a taxonomy but no method you can plug into your model.

Summary Table: Research Papers on Agentic AI, LLMs, Self-Improving Agents and IoT Service Composition (2020–2026)

S.No	Category	Paper	Year	Domain	Abstract
1	Service composition with LLMs and agents	Agentic Service-Oriented Computing: A Manifesto for the Next Frontier of Service-Oriented Computing	2026	Service-oriented computing, agentic AI	A vision paper arguing that LLM-based agents add a new kind of service composition. These agents can propose, adapt and run compositions at runtime, going beyond manual specification and formal planning. It lists open research questions on composition models, standardised agent service interfaces, QoS-aware composition and semantic interoperability. It also argues for combining LLM flexibility with formal verification, because LLM matching is probabilistic.
2	Service composition with LLMs and agents	The IoT Whisperer: A Framework for Intelligent IoT Service Composition through LLMs	2025	IoT, service composition, LLM	Manually configuring IoT workflows is slow and hard to scale. The paper uses the Web of Things framework, Node-RED and LLMs to turn high-level natural-language system descriptions into executable workflows. Workflow components are generated automatically, so non-experts can build IoT systems.
3	Service composition with LLMs and agents	Leveraging LLMs for Dynamic IoT Systems Generation through Mixed-Initiative Interaction	2025	IoT, LLM agents, dynamic service generation	An LLM-agent architecture for IoT systems that adapt at runtime. It has a service discovery step, and when no existing service matches a requirement, an LLM coding agent generates a new one. The new service is registered in the system's context and registry. A user and the system interact to refine goals.

S.No	Category	Paper	Year	Domain	Abstract
4	Service composition with LLMs and agents	FAME: QoS-aware Async Service Orchestration for Agentic Workflows	2026	Agentic workflows, QoS, serverless orchestration	Agents that call MCP tools face latency, cost and context-size problems when large tool outputs pass through prompts. The paper treats MCP service input and output as a first-class QoS concern. Its optimizations include serverless (FaaS) wrapping, S3 handle passing, output caching and async callbacks. It evaluates ReAct, Reflexion, LLM Compiler and MLZero workflow patterns.
5	LLM (in IoT)	Large Language Models in the IoT Ecosystem: A Survey on Security Challenges and Applications	2025	LLM, IoT, security (survey)	A systematic literature survey of 30 key papers, mostly from 2020 to mid-2024. It covers LLM applications in IoT (smart cities, healthcare, industrial IoT), along with security, privacy, latency and edge-computing challenges.
6	LLM (in IoT)	LLMind 2.0: Distributed IoT Automation with Natural Language M2M Communication and Lightweight LLM Agents	2025	LLM agents, IoT automation, distributed systems	Centralized LLM control of IoT devices does not scale to large heterogeneous systems. LLMind 2.0 has a central coordinator split a user's instruction into simple natural-language subtasks. Lightweight LLM agents on each device then generate the device-specific code locally. Tests on multi-robot warehouse and WiFi scenarios show better scalability, reliability and privacy than the centralized approach.
7	Agentic AI	Agentic AI: Architectures, Taxonomies, and Evaluation of LLM Agents	2026	Agentic AI, LLM agents (survey)	AI is moving from text generation to agents that perceive, reason, plan and act. The paper reviews agent architectures, from single-loop agents to hierarchical multi-agent systems. It proposes a taxonomy of Perception, Brain, Planning, Action, Tool Use and Collaboration. It lists open challenges such as hallucinated actions, infinite loops and prompt injection.

S.No	Category	Paper	Year	Domain	Abstract
8	Agentic AI	Agentic AI: A Comprehensive Survey of Architectures, Applications, and Future Directions	2025	Agentic AI, multi-agent systems (survey)	A systematic review of the shift from passive AI tools to autonomous systems that pursue goals. It argues that people often wrongly equate modern neural agentic systems with older symbolic ones, which it calls "conceptual retrofitting". It covers architectures, multi-agent collaboration, applications and governance.
9	Self-improvement in agents	Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents	2025 (ICLR 2026)	Self-improving agents, coding agents, evolutionary search	The Gödel machine was a theoretical AI that rewrites itself only when it can prove the change helps. The Darwin Gödel Machine instead validates each self-modification empirically on coding benchmarks. It keeps an archive of generated agents and picks parents from it to branch into new variants. SWE-bench performance rose from 20% to 50%, and Polyglot from 14.2% to 30.7%. It outperformed baselines without self-improvement or open-ended exploration.
10	Self-improvement in agents	A Survey on Self-Evolution of Large Language Models	2024	Self-evolving LLMs and agents (survey)	LLMs and agents increasingly learn from experience that they generate themselves. The survey gives a conceptual framework of experience acquisition, refinement, updating and evaluation. It categorizes evolution objectives and summarizes the literature for each module. It also covers related ideas such as self-instruct, self-play and self-training.